MILITARY SPECIFICATION

TRANSDUCER, MOTIONAL PICKUP TRU-164/A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a differential (airspeed)-absolute (altitude) pressure transducer for aircraft applications, designated TRU-164/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-M-7793	Meter, Time Totalizing
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-R-83165	Recorder, Signal Data MXU-553/A

STANDARDS

Federal

FED-STD-595	Colors

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Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS17321	Meter, Time Totalizing, Miniature, Digital 28 Volt DC
MS24265	Connector, Receptacle, Electric - Single Hole Mount, Miniature
MS33649	Boss, Fluid Connection-Internal Straight Thread

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Aeronautics and Space Administration

NASA-TN-D-822 (Aug 1961 issue)	Tables of Airspeed, Altitude and Mach Number

(Application for copies should be addressed to the National Aeronautics and Space Administration, 1520 H Street NW, Washington, DC 20325.)

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3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The transducer shall consist of the following major components:

Impact pressure transducer:
0 to 42.938 (Inches Hg Differential)

Static pressure transducer:
31.019 to 0.810 (Inches Hg absolute)

Electronics module

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to atmospheric and climatic conditions to which the transducer will be subjected during storage and normal service.

3.4.2.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transducer, except where magnetic materials are essential.

3.4.4 Nonferrous materials. Nonferrous materials shall be used for all parts of the transducer, except where ferrous materials are essential. Nonferrous materials contained within hermetically sealed enclosures shall be considered suitably protected from corrosion. Requirements specified herein for fungicidal and corrosion protective treatment and anodizing of aluminum alloy parts do not apply for parts within hermetically sealed enclosures.

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3.4.5 Toxic and corrosive fumes. Materials used in the construction of the transducer shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.4.6 Protective treatment. When materials are used in the construction of the transducer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The transducer shall be so designed and constructed that no part will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, and installation.

3.5.1 Size. The dimensional outline of the transducer shall be as shown on figure 1. The dimensions shall include shock mounts, if required, which will be an integral part of the transducer. (See 3.5.12.)

3.5.2 Interchangeability. The transducer shall consist of a single unit. All transducers of the same type shall be electrically and mechanically interchangeable so that any transducer may be used with the recording system specified in MIL-R-83165 without modification, calibration, or adjustment. Potentiometric transducers shall not be used. Transducer designs shall be approved by the procuring activity.

3.5.3 Electronic components. Electronic components and the application thereof shall be in accordance with MIL-E-5400.

3.5.4 Output signals. The transducer shall be designed to provide 0 to 5V dc output signals linear with static and impact pressure for operation with an airborne signal data recording system which measures flight parameters relating to aircraft structural loads.

3.5.5 Polarity

3.5.5.1 Airspeed. With increasing differential pressure, the output voltage of the transducer shall become more positive.

3.5.5.2 Altitude. With decreasing static pressure (increasing altitude), the output voltage of the transducer shall become more positive.

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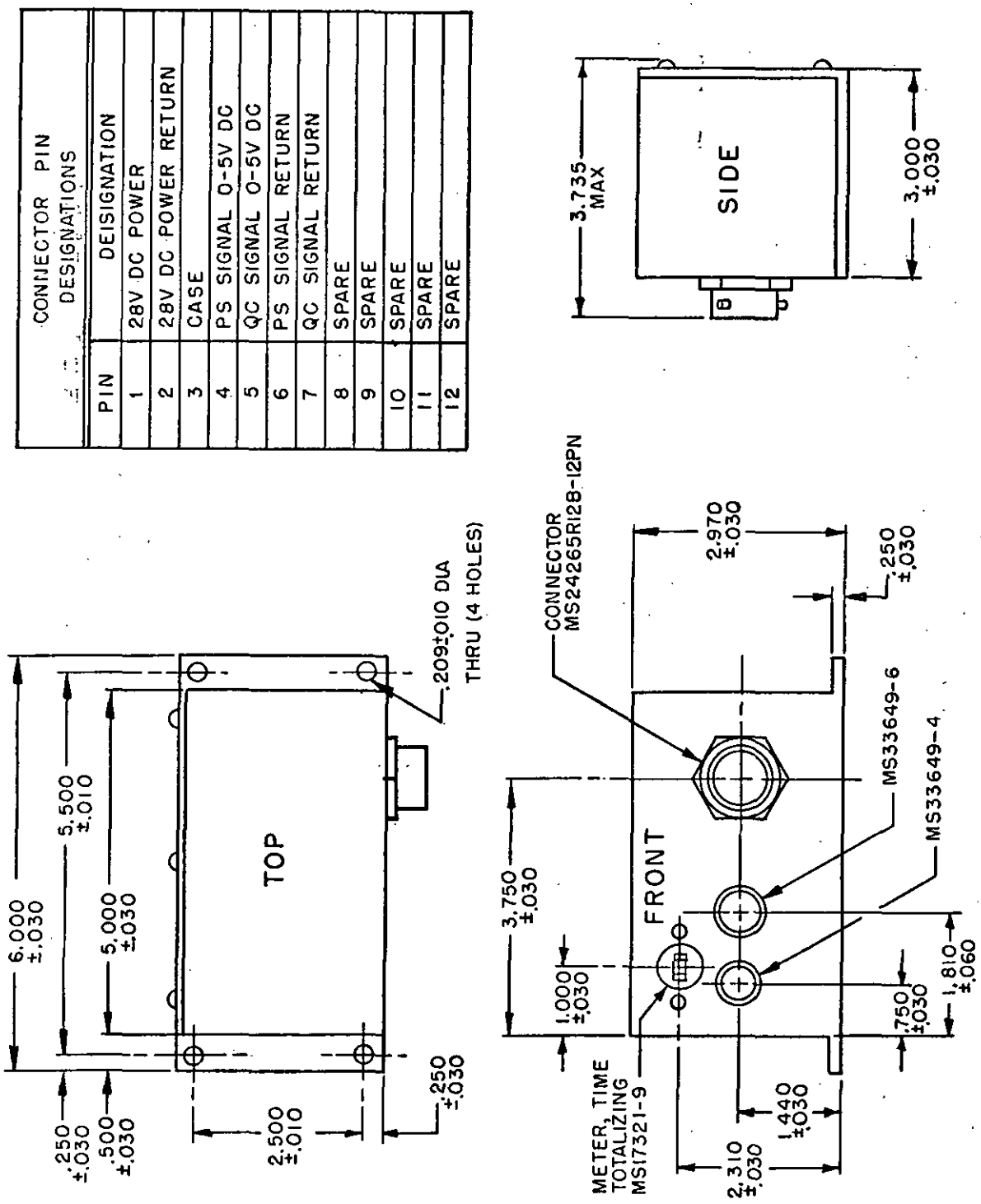


FIGURE 1. Transducer Dimensional Outline

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3.5.6 Power. The transducer shall be designed to operate from a 28V-dc power source as defined in MIL-STD-704.

3.5.6.1 Power variation. The transducer shall operate properly and continuously at voltages ranging from 22.5 to 30.0V dc and under the transient power conditions specified in MIL-STD-704.

3.5.7 Connector. The connector shall be an MS24265R12B12P with pin assignments as shown on figure 1.

3.5.8 Mounting. The transducer shall include an integral base plate for mounting. The configuration of the base plate shall be approved by the procuring activity.

3.5.9 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.5.9.1 Mean-time-between-failures. The transducer shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,000 hours at a 90 percent confidence factor.

3.5.9.1.1 Longevity. The transducer shall be designed for a service life of 3,000 hours with reasonable servicing and replacement of parts.

3.5.10 Shelf life. The transducer shall have a shelf life of 30 days.

3.5.11 Maintainability. The contractor shall establish a maintainability program in accordance with MIL-STD-470. The maintenance concept for this transducer is that maintenance shall be accomplished at depot level only.

3.5.11.1 Repair time. The transducer shall be completely repairable at depot level. The maintenance time for a typical repair shall not exceed 8 hours using one depot technician and authorized depot AGE and technical information. (Spare parts and facilities are assumed to be available.)

3.5.11.2 Accessibility. The transducer shall be so designed and constructed that interior parts, terminals, and wiring will be readily accessible to facilitate adjustments, circuit checking, calibration, and removal and replacement of parts. If hermetic sealing is utilized, this requirement shall apply after removal of the case.

3.5.12 Shock mounts. If required, shock mounts shall be provided and shall be within the envelope dimensions shown on figure 1. The design shall be such that the maximum sway space will be 1/4 inch in any direction.

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3.5.13 Inputs. The transducer shall be designed to receive total and static pressure inputs from the aircraft pitot-static system.

3.5.14 Pneumatic pressure connections. The pitot and static pressure fittings shall be located as shown on figure 1. The pitot connection shall be in accordance with MS33649-4, and the static connection shall be in accordance with MS33649-6.

3.5.14.1 Electrical connector. The electrical connector shall be as shown on figure 1 and shall be of the sealed, bayonet, pin-contact type.

3.5.15 Operational envelope. The transducer shall be capable of operating under any combinations of input pressures, temperatures, and voltages specified herein.

3.5.16 Electrical outputs. The electrical outputs of the transducer shall be 0 to 5V dc in accordance with table I and figure 2. The output signal commons shall be electrically isolated from the 28V dc power. The output impedances shall nominally be 7.5 ohms and shall not exceed 10 ohms.

TABLE I. Calibration Points

Static Pressure (In. Hg)	Output + DC (Volts)	Impact Pressure (In. Hg Differential)	Output + DC (Volts)
31.019	0.000	0.000	0.000
27.998	0.500	4.294	0.500
*24.977	1.000	*8.588	1.000
21.956	1.500	12.881	1.500
18.935	2.000	17.175	2.000
15.915	2.500	21.469	2.500
*12.894	3.000	*25.763	3.000
9.873	3.500	30.057	3.500
6.852	4.000	34.350	4.000
* 3.831	4.500	*38.644	4.500
0.810	5.000	42.938	5.000

* Denotes special test pressures referred to in section 4.

Note: Pressures per NASA-TN-D-822 (August 1961 issue)

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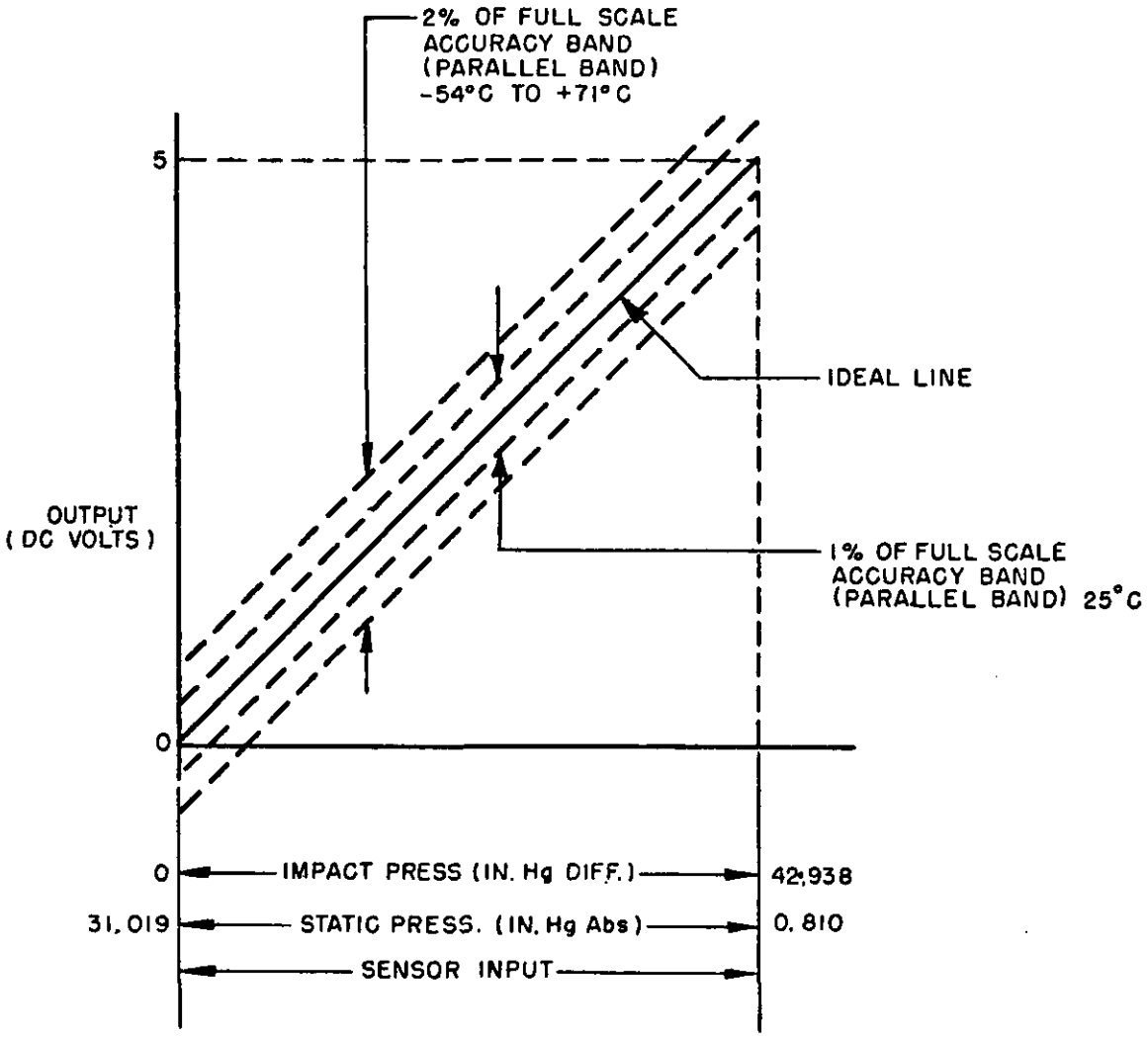


FIGURE 2. Calibration and Accuracy Curve

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3.5.16.1 Calibration. Calibration curves shall be provided with the transducer (see 6.4). Calibration points specified in table I shall be recorded in both ascending and descending directions. The output voltage of the transducer at each calibration point shall comply with 3.6.1.

3.5.17 Elapsed time indicator. A 10,000-hour digital-type elapsed time indicator conforming to MIL-M-7793 and MS17321-9 shall be provided in the transducer. The specific location of the indicator shall be as approved by the procuring activity.

3.5.18 Case. The case shall conform to the envelope dimensions shown on figure 1 and shall be finished in lusterless black, color 37038 of FED-STD-595. It shall provide a sealed enclosure for the internal mechanism and shall be of such design that the internal mechanism may be removed from the case, replaced, and the case resealed without use of special tools and fixtures unless they are approved by the procuring activity. Sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the transducer may be subjected. Seal effectiveness shall be determined by successful performance per 3.6.

3.5.18.1 The sealed case shall withstand an external pressure of 56 inches Hg absolute without failure of the seal or such distortion of the case as to affect subsequent normal operation of the transducer.

3.5.19 Lubrication. If used in the transducer, lubricants shall be such that satisfactory operation will be insured when the equipment is subjected to the environmental conditions specified herein. The transducer shall operate satisfactorily for 1,000 hours without requiring lubrication.

3.5.20 Threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.6 Performance. The transducer shall be capable of meeting the following without the use of internal or external vibrators, except as otherwise specified:

a. Continuous operation over a temperature range of -54° to +71°C (-65° to 160°F). The temperature shall remain constant for long periods of time and shall vary at rates up to and including 2°C (36°F) per second.

b. Storage temperatures ranging from -62° to +71°C (-80° to +160°F)

c. Relative humidity up to 95 percent.

d. Static pressure (In. Hg absolute): 31.019 to 0.810.

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e. Impact pressure (In. Hg diff.): 0 to 42.938.

f. Operation over the temperature-altitude ranges specified in MIL-STD-810, the temperature and altitude remaining constant or varying at rates as high as 80,000 fpm or 2°C (36°F) per second.

g. Exposure to moist salt-laden atmosphere.

h. Vibration up to ±10g at frequencies ranging from 5 to 500 Hz.

i. Fungus growth as encountered in tropical climates.

j. Airborne dust particles as encountered in desert operation.

k. Shock forces up to 15g operating and 20g nonoperating for 10 milliseconds.

l. Acceleration force of 20g for 30 seconds.

3.6.1 Accuracy. The maximum allowable deviation of any point, from an ideal line on a calibration plot shall be within ±1 percent of full scale for standard atmosphere conditions. The accuracy at the extreme environmental conditions shall be ±2.0 percent, increasing with the environment. (See table I for calibration points and figure 2 for the calibration plot. This is the maximum permissible error detectable in the output of the pressure transducer caused by individual parameters or simultaneously interacting parameters which shall include linearity, repeatability, hysteresis, resolution, noise, temperature, vibration, other environmental effects, and any other causes of error attributable to the pressure transducer.) This requirement shall apply to repeated exposure to any environmental conditions or combination of environmental conditions specified herein throughout the life of the transducer.

3.6.2 Repeatability. Under identical test and input signal conditions, the absolute output and differential output shall be repeatable to within 0.2 percent of full scale.

3.6.3 Hysteresis. Hysteresis error of the transducer shall be ±0.2 percent maximum of full scale.

3.6.4 Frequency response. The transducer frequency response shall be flat within 3 db to 5 Hz minimum.

3.6.5 Resolution. Resolution for the transducer shall be ±0.1 percent of full scale or better.

3.6.6 Position sensitivity. Position error of the transducer shall be ±0.06 percent maximum of full scale pressure.

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3.6.7 Overpressure. The transducer shall withstand an overpressure of at least 25 percent with no damage or degradation of performance.

3.6.8 Dielectric strength. The transducer shall withstand 100V dc applied between any connector pin (except pin 3, case ground) and case ground for 1 minute. The minimum resistance shall be 20 megohms.

3.6.9 Leakage. The change in output voltages shall not exceed an amount equal to 0.027 inch Hg/minute when an absolute pressure of 5.0 inches Hg is applied to the static pressure connection or 55 inches Hg is applied to the total pressure connection of the transducer.

3.7 Electromagnetic interference. The transducer shall meet the electromagnetic interference requirements of MIL-STD-461 as specified for class A1 equipment.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Weight. The weight of the transducer shall not exceed 3.5 pounds.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.11 Finishes and protective coatings. Finishes and protective coatings shall be as follows, unless protected by a sealed enclosure:

3.11.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.11.1.1 Aluminim alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.11.1.2 Where the primary purpose of the treatment is to affort a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.11.1.3 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.11.1.4 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

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3.11.2 Cadmium plating. Cadimum plating shall be in accordance with QQ-P-416 and of a class that is adequate to achieve the degree of protection required.

3.12 Workmanship. The transducer shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine-screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

3.12.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the transducer, they shall be held or limited accordingly.

3.12.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt threads.

3.12.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.12.4 Cleaning. The transducer shall be thoroughly cleaned and loose, spattered, or excess solder; metal chips; or other foreign material removed after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the transducer shall be classified as follows:

a. Preproduction testing

b. Quality conformance tests.

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4.3 Test conditions

4.3.1 Standard atmospheric conditions. When the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg), at room temperature, approximately 25°C (77°F), and with a relative humidity of not more than 90 percent. When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Reference standard barometer. The reference standard for atmospheric pressure shall be a mercury barometer, maintained in accordance with the manufacturer's recommendations, which, with correction, shall be accurate to within ±0.005 inch Hg. If the barometer used is new in service, it should be checked approximately every 6 months until the stability of its calibration is established.

4.3.3 Warmup. Unless otherwise specified (see 6.2), the transducer shall be in operating condition and performing in accordance with the requirements of this specification within the following warmup periods:

a. Two minutes after power has been applied if the transducer has been at a temperature of not less than -55°C (67°F) prior to excitation

b. One minute after power has been applied if the transducer has been at a temperature of not less than -20°C (-40°F) prior to excitation.

4.3.4 Attitude. Unless otherwise specified, the transducer shall be tested in its normal operating position.

4.3.5 Test equipment. During testing, the pressure connections shall be connected to precision liquid barometers and manometers (or equivalent), the accuracy of which has been certified by comparison with primary standards. Certification of the prime standard shall be current and in no case shall the certification interval be longer than 6 months.

4.3.6 External connections. When external connections are required to simulate installed conditions, standard AN connectors and at least 10 feet of copper or aluminum tubing, coiled and so arranged as to allow drainage of condensed vapors, shall be utilized and attached to the pitot and static pressure ports of the transducer.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of nine transducers representative of the production equipment. Three of the samples shall be subjected to all the tests except reliability and shelf life, and the other six samples

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shall be subjected to the reliability and shelf life tests only. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance tests. The quality conformance tests shall consist of the following:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each transducer shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Scale error test

c. Hysteresis test

d. Repeatability test

e. Resolution test

f. Response test

g. Leakage tests

h. Position error test

i. Friction error test.

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. One transducer shall be selected at random from each one hundred, or less, produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Low temperature operation test

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c. High temperature operation test

d. Vibration error test

e. Dielectric strength tests.

4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), 3 transducers shall be selected at random from the first 20 on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling Plan A tests

b. Voltage variation test

c. Electromagnetic interference test

d. Vibration failure test

e. Low temperature exposure test

f. High temperature exposure test

g. Humidity test

h. Dust test

i. Fungus test

j. Salt-fog test

k. Acceleration test

l. Mechanical shock test

m. Overpressure test.

4.5.2.3 Rejection and retest. When a transducer selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extent and cause of failure are determined. Corrections made shall be approved by the procuring activity. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But acceptance of transducers on hand or later produced shall not be made until it is determined that items meet all requirements of the specification.

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4.5.2.4 Sampling plan C. Unless otherwise specified (see 6.2), 6 additional transducers shall be selected at random from the first 20 on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Reliability.

4.5.2.4.1 Rejection or acceptance. Rejection or acceptance shall be determined as specified in 4.6.26.2.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in transducers already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The transducer shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Dielectric strength tests. A voltage of 100V dc shall be applied between all terminals and the case (except pin 3). In addition 100V dc shall be applied between pins 1 to pins 3, 4, 5, 6, and 7; and between pins 2 to pins 3, 4, and 7; and pin 5 to pins 3, 4, and 6. In no case shall the resistance be less than 20 megohms and no damage shall result from this test.

4.6.3 Scale error test. The transducer shall be tested for scale errors in both ascending and descending directions at the test points specified in table I. The outputs shall not exceed the tolerances specified in 3.6.1 and as shown on figure 2.

4.6.4 Hysteresis test. Not more than 15 minutes after the transducer has been subjected to the upper limits of the scale error test (4.6.3) in an ascending (altitude or differential pressure) direction, it shall be subjected to a scale error test as specified in 4.6.3 at the asterisked test points specified in table I in a descending direction. The difference between outputs in the ascending and descending directions shall not exceed 0.010V dc.

4.6.5 Repeatability test. Readings shall be taken at the asterisked test points specified in table I. Two more sets of readings shall be taken under the same environmental conditions. No reading shall differ from the average of the three readings by more than 0.010V dc.

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4.6.6 Resolution test. The transducer shall be tested at the asterisked test points specified in table I. A change of static pressure of ±0.031 inch Hg shall produce a change in the static pressure output voltage. A change of differential pressure of 0.043 inch Hg shall produce a change in the differential pressure output voltage.

4.6.7 Response test. A 1-psi pressure step shall be applied to the static pressure connection, and the static and differential pressure output voltages shall be monitored with an oscilloscope or recording oscilloscope. The output signals shall reach 63 percent of the final values in 32 milliseconds, or less.

4.6.8 Leakage tests. The transducer shall be subjected to the following leakage tests:

4.6.8.1 Static pressure test. A pressure of 5 inches Hg absolute shall be applied to the static pressure connection. The connection tubing shall then be pinched off or otherwise completely sealed within 2 inches of the input pressure port of the transducer, and the output voltages monitored for 1 minute. The change in output voltages after the 1-minute period shall not exceed 0.0045V dc (0.027 inch Hg) for static pressure nor 0.0031V dc for differential pressure.

4.6.8.2 Total pressure test. A pressure of 55 inches Hg absolute shall be applied to the total pressure connection, and the static pressure shall be vented. The connection tubing shall be pinched off or otherwise completely sealed within 2 inches of the input pressure port on the transducer, and the output monitored for 1 minute. The change in output voltage after the 1-minute period shall not exceed 0.0031V dc (0.027 inch Hg).

4.6.9 Position error test. The transducer shall be rotated 90° in four different directions from its normal mounting position to test position error. The change in output voltages shall not exceed 0.003V dc.

4.6.10 Friction error test. The transducer shall be tested for friction error at the specified asterisked test points of table I which shall be set without external vibration. External vibration shall then be applied. The external vibrations shall be applied by means of electromechanical vibrators. The vibration shall be approximately 0.2g with a frequency not less than 30 Hz. The readings shall not change by more than 0.033V dc for static pressure and 0.023V dc for differential pressure. This test may be combined with the test specified in 4.6.3.

4.6.11 Low temperature operation test. The transducer shall be placed in a test chamber and maintained at a temperature of -35° ±2°C (-31° ±4°F) for a period of 4 hours with no power applied. While the equipment is still at this temperature, power shall be applied for 2 minutes and the transducer tested for scale error at the test points specified in table I. The chamber temperature shall then be lowered to and maintained at -54° ±2°C (-65° ±4°F) for a period of 4 hours with no power applied to the transducer. While the equipment is still

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at this temperature, power shall be applied for 2 minutes and the transducer again tested for scale error at the test points specified in table I. While at both -35° and -54°C (-31° and -65°F), the transducer shall pass the leakage tests specified under 4.6.8.

4.6.12 High temperature operation test. The transducer shall be placed in a test chamber and maintained at a temperature of 71° ±2°C (160° ±4°F) for a period of not less than 4 hours with no power applied. While still at this temperature, power shall be applied for a period of 2 minutes and the transducer tested for scale error at the test points specified in table I. While at the 71°C (160°F) temperature, the transducer shall pass the leakage tests specified under 4.6.8.

4.6.13 Vibration error test. The transducer shall be subjected to a vibration cycling test at room temperature in accordance with Method 514.1, procedure I of MIL-STD-810, equipment category (b), curve AT. The transducer shall be tested during vibration for scale error at the test points specified in table I. During vibration, the tolerance shall be 50 percent greater than that specified for room temperature scale error. After this test, the tolerance shall be as specified for room temperature scale error.

4.6.14 Voltage variation test. The transducer shall be tested for accuracy at the test points specified in table I at 22.5 and 30V dc and under the transient conditions as specified in MIL-STD-704. The change in scale error shall not exceed 0.005V dc.

4.6.15 Electromagnetic interference test. The transducer shall be subjected to an electromagentic interference test in accordance with MIL-STD-462 as specified for class A1 equipment.

4.6.16 Vibration failure test. The transducer shall be subjected to a vibration test at room temperature in accordance with Method 514.1 of MIL-STD-810, equipment category (b) procedure I, curve AT. The transducer shall then be tested for scale error at the test points specified in table I. During vibration this unit shall be energized but not operated. After completion of this test, the unit shall be tested and the tolerance shall be as specified for room temperature scale error.

4.6.17 Low temperature exposure test. The transducer shall be placed in a temperature chamber and maintained at a temperature of -62° ±2°C (-80° ±4°F) for a period of not less than 48 hours. At the conclusion of this period, the transducer shall be returned to room temperature and subjected to the scale error test at the test points specified in table I.

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4.6.18 High temperature exposure test. The transducer shall be placed in a temperature chamber and maintained at a temperature of 85° ±2°C (185° ±4°F) for a period of not less than 48 hours. At the end of this period, the transducer shall be returned to room temperature and subjected to the scale error test at the test points specified in table I.

4.6.19 Humidity test. The transducer shall be subjected to a humidity test in accordance with Method 507, procedure I of MIL-STD-810. At the conclusion of this test, the transducer shall pass the scale error test at the test points specified in table I and the leakage tests specified under 4.6.8.

4.6.20 Dust test. The transducer shall be subjected to a dust test in accordance with Method 510, procedure I of MIL-STD-810. At the conclusion of this test, the transducer shall pass the scale error tests at the test points specified in table I and the leakage tests specified under 4.6.8.

4.6.21 Fungus test. The transducer shall be subjected to a fungus test in accordance with Method 508 of MIL-STD-810.

4.6.22 Salt-fog test. The transducer shall be subjected to salt-fog in accordance with Method 509 of MIL-STD-810. At the conclusion of this test, the transducer shall pass the scale error tests at the test points specified in table I and the leakage tests specified under 4.6.8.

4.6.23 Acceleration test. The transducer shall be subjected to an acceleration test in accordance with MIL-STD-810, Method 513.1, procedure I, except the acceleration shall be 20g for 30 seconds. Before and after this test, the transducer shall be operated and tested for scale error at the test points specified in table I.

4.6.24 Mechanical shock test. The transducer shall be subjected to a shock test in accordance with Method 516.1, procedure I, of MIL-STD-810. At the conclusion of this test, the transducer shall pass the scale error test at the test points specified in table I and the leakage tests specified under 4.6.8.

4.6.25 Overpressure test. Static pressures equal to 0.31 and 39 inches Hg shall be applied alternately to the transducer. Differential pressures of -10 and +52 inches Hg shall then be applied alternately. There shall be no damage to the internal mechanism due to this pressure and no calibration shift shall result. At the conclusion of this test, the transducer shall pass the scale error test of 4.6.3.

4.6.26 Reliability test. A minimum of six transducers shall be subjected to a reliability test in accordance with the following:

4.6.26.1 Reliability cycling. Each transducer shall be subjected to a minimum of three pressure cycles per hour for a minimum of 880 hours or 2640 cycles consisting of a pressure variation equivalent to approximately three-fourths

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of full scale. Each pressure cycle shall consist of operation to maximum pressure and return to ambient. The maximum scale value shall be maintained for not less than 2 minutes. Approximately 2 minutes at ambient pressure shall be allowed between each cycle. During the fifth cycling period, the transducer shall be vibrated at an amplitude of 0.009 to 0.11 inch and a frequency of 35 Hz. The cycling period, number of hours or cycles, and cycling temperature shall be accomplished in accordance with table II. After each period of cycling, the transducer shall be subjected to and shall meet the room temperature scale error test, except that only six test stations, representative of the low range, midrange, and high range shall be tested. Figure 2 tolerances shall apply. During each scale error test, when any two test points exceed allowable tolerances by 0.07 percent FS, one malfunction shall be charged against this test. Upon completion of the final cycle, the transducer shall be rested at ambient pressure for a period of 48 hours and then subjected to the individual tests. Failure to meet any individual test shall be considered a malfunction. For each five test points out of tolerance during the scale error test, one malfunction shall be charged. In case of a malfunction, the malfunction shall be recorded and the test interrupted while repairs are made. The test shall then be continued from the point of interruption. If recalibration is required to pass the scale error test, an additional malfunction shall be charged against this test.

TABLE II. Reliability Cycling

Cycling Period	Hours or Cycles		Ambient Temperature °C
1	150	450	25
2	100	300	55
3	150	450	25
4	100	300	-35
5	100	300	25
6	100	300	55
7	100	300	-35
8	100	300	25

4.6.26.2 MTBF. The reliability test shall demonstrate a minimum acceptable MTBF of 1,000 hours to a 90 percent confidence level which will be verified when the six transducers have met the conditions of table III.

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TABLE III. Acceptance or Rejection

M (Maximum Number of Failures Corresponding to the Specified T)	T (Total Test Time) Hours or Cycles	
0	5,280	15,840
1	6,240	18,720
2	7,200	21,600
3	8,160	24,480
4	9,120	27,360
5	10,200	30,600

4.6.26.3 Longevity test. Unless otherwise specified (see 6.2), two out of every six transducers tested for reliability shall be subjected to a longevity test in accordance with MIL-STD-781 until a total of 3,000 hours on each is attained.

4.6.26.4 Shelf life test. After completion of the reliability test, the remaining four of the six transducers specified in 4.6.26.3 shall be placed in storage for a period of not less than 30 days. During the 30-day period, the transducer shall remain as dormant as practicable. The mechanism shall not be exercised other than that caused by normal ambient pressure vibrations. The temperature shall remain essentially constant; however, normal room temperature variations will be permissible. After the 30-day period, the transducer shall again meet the individual tests.

4.6.27 Maintainability demonstration. A maintainability demonstration test shall be conducted in accordance with MIL-STD-471 on a minimum of three selected repairs. The tasks shall be selected in advance of timing the tasks and shall be approved by the procuring activity.

4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Transducers shall be preserved and packaged in accordance with MIL-STD-794 and method III of MIL-P-116, one each in a unit container conforming to PPP-B-636, domestic class. Electrical receptacles and connectors shall be equipped with water-vaporproof caps.

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5.1.2 Level C. Transducers shall be individually preserved and packaged in a manner that will provide protection against corrosion, deterioration, or physical damage so that serviceability is assured during shipment from supply source to the first receiving activity. This level may be the contractor's commercial practice provided the latter meets the criteria for this level as defined in MIL-STD-794.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Transducers preserved and packaged as specified in 5.1.1 shall be packed as required for consolidation in shipping boxes conforming to PPP-B-636, weather-resistant class. Exterior shipping containers shall be of uniform shape and size, of minimum tare, weight, and cube.

5.2.2 Level B. Transducers preserved and packaged as specified in 5.1.1 shall be packed as required for consolidation in PPP-B-636, domestic class exterior shipping containers. Other requirements as specified in 5.2.1 shall apply.

5.2.3 Level C. Transducers shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery, and shall meet as a minimum, the requirements of carrier rules and regulations. This level may be the contractor's commercial practice provided the latter meets the criteria for this level as defined in MIL-STD-794.

5.3 Marking. In addition to the standard marking requirements of MIL-STD-129, level C/C packs containing item(s)/equipment intended for immediate use at destination shall be conspicuously marked as follows on two opposite sides of the exterior containers in letters as large as space permits:

FOR IMMEDIATE USE
DO NOT STORE IN THIS PACK

5.3.1 Nomenclature. The nomenclature shall be as follows:

TRANSDUCER, MOTIONAL PICKUP TRU-164/A

6. NOTES

6.1 Intended use. The TRU-164/A transducer is intended for use in providing altitude and airspeed information for operation with an airborne recording set.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When the warmup periods specified in 4.3.3 are not applicable

c. When sampling plan B tests are not to be conducted (see 4.5.2.2)

d. When sampling plan C tests are not to be conducted (see 4.5.2.4)

e. If longevity test is to be other than as specified in 4.6.5.3

f. Required levels of preservation and packaging, and packing (see section 5).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Scale error. Scale error is the error in transducer indication due to the physical and electrical properties, construction, and adjustment of the aneroid and compensation system.

6.3.2 Hysteresis. Hysteresis is the error in transducer indication introduced during an increase or decrease in height due to imperfect elastic properties of the aneroid material which prevents the aneroid from assuming its normal shape for any given atmospheric pressure.

6.3.3 Friction error. Friction error is the error in transducer indication due to friction in the mechanism.

6.3.4 Temperature error. Temperature error is the error in transducer indication due to the effect of temperature variation on the transducer.

6.3.5 Position error. Position error is the error in transducer indication due to a change in the state of static balance of the mechanism when the instrument is rotated from the test position to other positions.

6.3.6 Tolerance. Tolerance is the maximum permissible deviation from a specified value for a particular test. A positive tolerance is the maximum permissible deviation above the specified value and a negative tolerance is the maximum permissible deviation below the specified value.

6.3.7 Standard atmospheric conditions. Standard atmospheric conditions refer to those normally found in a calibration test area.

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6.4 Data. Data generated by this specification is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) referencing the appropriate data item description in the military departments' Authorized Data List (ADL). The data produced by this document is as follows:

a. Two copies of the calibration curves specified in 3.5.16.1

b. The test report specified in 4.4.2.

Custodian:
Air Force - 11

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